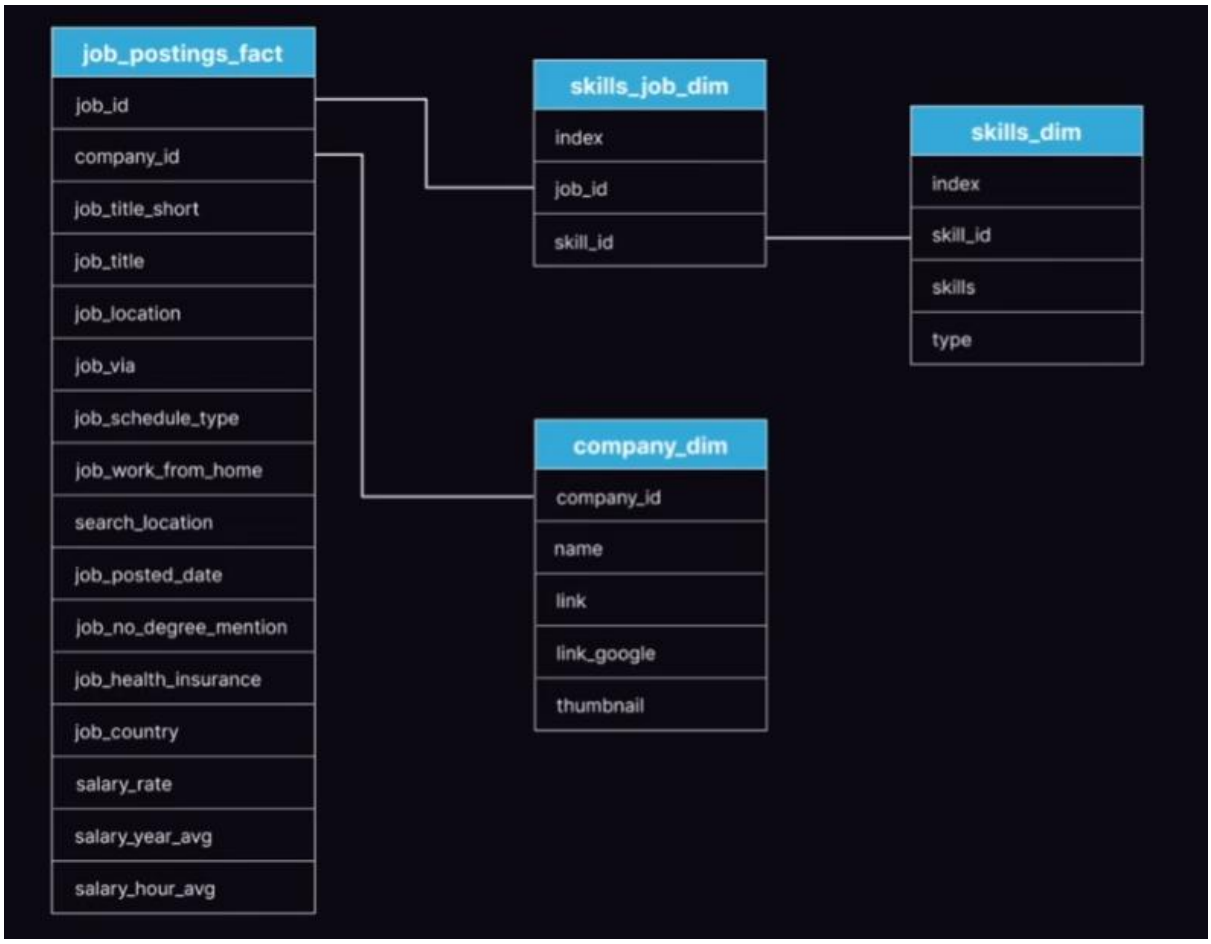
1. Project Overview

This project analyses the jobs available in the market (dummy data) and analyse the top-paying roles and skills. The goal is to uncover insights based on different job types, skills required, average salary and various other factors to identify the best fit role in the market to be targeted.

2. Dataset Summary

- The dataset consists of 4 tables and the relationship is given as follows in the image.



3.Tools Used:

SQL: For querying the database and unearth critical insights.

PostgreSQL: database management system, ideal for handling the job posting data.

Visual Studio Code: For coding and executing SQL queries.

Git & GitHub: For version control and sharing SQL scripts and analysis.

4. Data Analysis using SQL:

1. What are the top-paying data analyst jobs?

job_id	job_location	job_title	job_schedule_type	salary_year_avg	job
Filter...	Filter...	Filter...	Filter...	Filter...	
226942	Anywhere	Data Analyst	Full-time	650000.0	20
547382	Anywhere	Director of Analytics	Full-time	336500.0	20
552322	Anywhere	Associate Director- Data Insights	Full-time	255829.5	20
99305	Anywhere	Data Analyst, Marketing	Full-time	232423.0	20
1021647	Anywhere	Data Analyst (Hybrid/Remote)	Full-time	217000.0	20
168310	Anywhere	Principal Data Analyst (Remote)	Full-time	205000.0	20
731368	Anywhere	Director, Data Analyst - HYBRID	Full-time	189309.0	20
310660	Anywhere	Principal Data Analyst, AV Performance...	Full-time	189000.0	20
1749593	Anywhere	Principal Data Analyst	Full-time	186000.0	20
387860	Anywhere	ERM Data Analyst	Full-time	184000.0	20
1781684	Anywhere	DTCC Data Analyst	Full-time	170000.0	20
987523	Anywhere	Azure Data Python Consultant - contra...	Contractor	170000.0	20
1352513	Anywhere	REMOTE Director of Data Analytics	Full-time	170000.0	20
1525451	Anywhere	Manager, Data Analyst	Full-time	167000.0	20
712473	Anywhere	Data Analyst	Full-time	165000.0	20
1246069	Anywhere	Data Analyst	Full-time	165000.0	20
1799883	Anywhere	Principal Data Science Analyst- Remote	Full-time	164746.0	20

2. What skills are required for these top-paying jobs?

job_id	job_title	salary_year_avg	company_name	skills
Filter...	Filter...	Filter...	Filter...	Filter...
552322	Associate Director- Data Insights	255829.5	AT&T	sql
552322	Associate Director- Data Insights	255829.5	AT&T	python
552322	Associate Director- Data Insights	255829.5	AT&T	r
552322	Associate Director- Data Insights	255829.5	AT&T	azure
552322	Associate Director- Data Insights	255829.5	AT&T	databricks
552322	Associate Director- Data Insights	255829.5	AT&T	aws
552322	Associate Director- Data Insights	255829.5	AT&T	pandas
552322	Associate Director- Data Insights	255829.5	AT&T	pyspark
552322	Associate Director- Data Insights	255829.5	AT&T	jupyter
552322	Associate Director- Data Insights	255829.5	AT&T	excel
552322	Associate Director- Data Insights	255829.5	AT&T	tableau
552322	Associate Director- Data Insights	255829.5	AT&T	power bi
552322	Associate Director- Data Insights	255829.5	AT&T	powerpoint
99305	Data Analyst, Marketing	232423.0	Pinterest Job Advertis...	sql
99305	Data Analyst, Marketing	232423.0	Pinterest Job Advertis...	python
99305	Data Analyst, Marketing	232423.0	Pinterest Job Advertis...	r
99305	Data Analyst, Marketing	232423.0	Pinterest Job Advertis...	hadoop

3. What skills are most in demand for dataana lysts?

sql	385750
python	381863
aws	145718
azure	132851
r	131285
tableau	127500
excel	127341
spark	114928
power bi	98363
java	85854
sas	83652
hadoop	65041
scala	57058
snowflake	55150
databricks	53282
gcp	50824
git	50358
kafka	50092
airflow	45363
oracle	45318

4. Which skills are associated with higher salaries?

skills	avg_salary
Filter...	Filter...
pyspark	208172
bitbucket	189155
watson	160515
couchbase	160515
datarobot	155486
gitlab	154500
swift	153750
jupyter	152777
pandas	151821
elasticsearch	145000
golang	145000
numpy	143513
databricks	141907
linux	136508
kubernetes	132500
atlassian	131162

5. What are the most optimal skills to learn?

skill_id	skills	demand_count	avg_salary
8	go	27	115320
234	confluence	11	114210
97	hadoop	22	113193
80	snowflake	37	112948
74	azure	34	111225
77	bigquery	13	109654
76	aws	32	108317
4	java	17	106906
194	ssis	12	106683
233	jira	20	104918
79	oracle	37	104534
185	looker	49	103795
2	nosql	13	101414
1	python	236	101397
5	r	148	100499
78	redshift	16	99936
187	qlik	13	99631

5. Key Insights:

1. Top-Paying Data Analyst Jobs: The highest-paying jobs for data analysts that allow remote work offer a wide range of salaries, the highest at $650,000!

2. Skills for Top-Paying Jobs: High-paying data analyst jobs require advanced proficiency in SQL, suggesting it's a critical skill for earning a top salary.

3. Most In-Demand Skills: SQL is also the most demanded skill in the data analyst job market, thus making it essential for job seekers.

4. Skills with Higher Salaries: Specialized skills, such as SVN and Solidity, are associated with the highest average salaries, indicating a premium on niche expertise.

5. Optimal Skills for Job Market Value: SQL leads in demand and offers fgr a high average salary, positioning it as one of the most optimal skills for data analysts to learn to maximize their market value.